

1. DATA ANALYST - E-Commerce Customer Analytics Platform

Problem Context

You're joining the analytics team at **ShopFast**, a mid-sized e-commerce platform with 2M monthly active users. The company is experiencing declining repeat purchase rates and the executive team needs to understand why. They've noticed that customer behavior differs significantly across acquisition channels, product categories, and geographic regions.

The marketing team recently ran a promotional campaign offering 20% discounts, but early indicators suggest it may have attracted low-quality traffic. The product team is also testing a new recommendation engine that was rolled out to 30% of users in March 2024.

Your task is to analyze 6 months of transactional data, identify patterns in customer behavior, and provide actionable recommendations to improve retention and lifetime value.

Dataset / Inputs

Datasets intentionally reflect real-world schema and documentation inconsistencies. Candidates are expected to reconcile differences and document assumptions as part of the analysis.

You'll work with **4 CSV files** containing messy, real-world data:

1. **customers.csv** (150,000 rows)

customer_id, signup_date, acquisition_channel, country, first_purchase_date, email_domain
100001, 2023-10-15, facebook_ads, US, 2023-10-16, gmail.com
100002, 2023-10-15, organic_search, UK, , yahoo.com
100003, 2023-10-16, instagram_ads, US, 2023-10-20, gmail.com
...

- Contains nulls in **first_purchase_date** (non-converters)
- **acquisition_channel** has inconsistent naming: "Facebook Ads", "facebook_ads", "FB_ADS"
- Some **signup_date** values are in the future (data quality issues)

2. **transactions.csv** (800,000 rows)

transaction_id, customer_id, transaction_date, product_id, quantity, unit_price, discount_applied, payment_method
T-50001, 100001, 2023-10-16 14:23:00, P-789, 2, 29.99, 0, credit_card
T-50002, 100001, 2023-11-02 09:15:00, P-456, 1, 89.99, 20, paypal

T-50003, 100004, 2023-10-17 18:45:00, P-123, 1, 15.50, 0, credit_card

...

- `discount_applied` is percentage (0-100)
- Some transactions have `quantity` = 0 or negative values (returns/errors)
- `transaction_date` spans Oct 2023 - Mar 2024
- Payment failures are marked with `unit_price` = 0

3. `products.csv` (5,000 rows)

product_id, category, subcategory, brand, base_price, cost_to_company, is_active

P-123, Electronics, Headphones, AudioTech, 15.50, 8.20, TRUE

P-456, Home & Garden, Furniture, ComfortCo, 89.99, 45.00, TRUE

P-789, Fashion, Shoes, TrendyFeet, 29.99, 12.50, FALSE

...

- Some products have been discontinued (`is_active` = FALSE)
- `base_price` doesn't always match `unit_price` in transactions (dynamic pricing)

4. `recommendation_exposure.csv` (400,000 rows)

customer_id, exposure_date, variant, products_shown, products_clicked

100005, 2024-03-01, control, "P-123,P-456,P-789", "P-456"

100006, 2024-03-01, treatment, "P-234,P-567,P-890", ""

100007, 2024-03-02, treatment, "P-111,P-222,P-333", "P-111,P-333"

...

- `variant` is either "control" (old system) or "treatment" (new recommendation engine)
- `products_clicked` can be empty string (no clicks)
- Same customer may appear multiple times with different dates

Core Tasks to Implement

Task 1: Data Quality & ETL

Create a robust data cleaning pipeline that:

- Identifies and documents all data quality issues
- Standardizes `acquisition_channel` values
- Handles invalid dates, negative quantities, and zero-price transactions
- Creates a master customer table with derived metrics:
 - `days_to_first_purchase` (signup → first purchase)
 - `total_transactions`

- `total_revenue`
- `total_discount_received`
- `avg_order_value`
- `is_repeat_customer` (>1 purchase)
- `customer_lifetime_days` (signup → last purchase)
- `recency_days` (last purchase → March 31, 2024)

Provide a data quality report showing:

- Missing value percentages per column
- Invalid record counts with examples
- Decisions made for each data issue

Task 2: Customer Segmentation Analysis

Perform RFM (Recency, Frequency, Monetary) segmentation:

- Calculate RFM scores for each customer (scale 1-5)
- Create 5 distinct customer segments with business-friendly names
- Analyze segment composition by:
 - Acquisition channel
 - Product category preference
 - Geographic distribution
 - Average discount dependency

Key questions to answer:

1. Which acquisition channels bring the highest-value customers?
2. Are discount-acquired customers less loyal?
3. What's the relationship between first-purchase timing and lifetime value?

Task 3: Cohort Retention Analysis

Build monthly signup cohorts (Oct 2023 - Mar 2024):

- Calculate retention rates at 30, 60, 90, 120, 150, 180 days
- Visualize cohort retention curves
- Identify which cohorts are performing best/worst
- Analyze if the March 2024 cohort shows anomalies (promotional campaign impact)

Calculate and compare:

- Cohort-specific AOV (average order value) trends
- Repeat purchase rates by cohort
- Revenue per cohort over time

Task 4: A/B Test Evaluation

Evaluate the recommendation engine experiment:

- Calculate CTR (click-through rate) for control vs. treatment
- Measure conversion impact (exposure → purchase within 7 days)
- Analyze revenue impact per user group
- Test statistical significance (use appropriate test)
- Segment analysis: Does the treatment effect vary by customer segment?

Provide recommendation: Should the company roll out the new engine to 100%?

Task 5: Business Insights & Dashboard

Create an executive dashboard (SQL + visualization) showing:

- KPI summary: Total revenue, active customers, avg order value, repeat rate
- Monthly trends (Oct 2023 - Mar 2024)
- Channel performance comparison
- Top 10 products by revenue and margin
- Geographic performance

Write a 2-page executive summary answering:

1. Why is repeat purchase rate declining?
2. Which customer segments should we focus on?
3. Is our promotional strategy working?
4. Should we invest more in specific acquisition channels?
5. What's the projected 6-month revenue if trends continue?

Technical Requirements

Mandatory Tools:

- **SQL:** All core analysis must be written in SQL (PostgreSQL, MySQL, or SQLite)
- **Python/R:** For statistical tests, visualization, and data cleaning
- **Visualization:** Tableau, Matplotlib/Seaborn, or Plotly
- **Documentation:** Jupyter Notebook or R Markdown showing full workflow

Performance Expectations:

- SQL queries must execute in <30 seconds on the full dataset
- Use appropriate indexes, CTEs, and window functions
- Avoid redundant scans; optimize for readability AND performance

Edge Cases to Handle:

- Customers who signed up but never purchased
- Transactions with returns (negative quantities)
- Product price changes over time
- Multiple transactions on the same day
- Customers in the recommendation experiment who didn't purchase

Scalability Considerations:

- Queries should work if dataset grows to 10M transactions
- Segment definitions should be parameterized (easy to adjust thresholds)
- Dashboard should auto-update with new data

Evaluation Criteria

Code Quality (25%)

- SQL style: readable CTEs, meaningful aliases, comments
- Python: modular functions, not monolithic scripts
- Variable naming and code organization

Analytical Rigor (30%)

- Correct statistical methods (not just descriptive stats)
- Appropriate hypothesis testing for A/B test
- Awareness of confounding variables (e.g., seasonality)
- Handling of edge cases and outliers

Business Acumen (25%)

- Insights are actionable, not just observations
- Recommendations are prioritized and justified
- Understanding of e-commerce metrics (CAC, LTV, churn)
- Executive summary is clear and non-technical

Data Quality (10%)

- Thorough data validation
- Documentation of cleaning decisions
- No silent data drops without justification

Visualization (10%)

- Charts are appropriate for the data type
- Clear labels, legends, titles
- Dashboard is intuitive and tells a story

Expected Deliverables

submission/

— README.md	# Setup instructions, key findings summary
— data_quality_report.md	# Issues found + decisions made
— sql/	
— 01_data_cleaning.sql	# ETL and cleaning logic
— 02_customer_master.sql	# Customer metrics table creation
— 03_rfm_segmentation.sql	# Segmentation logic
— 04_cohort_analysis.sql	# Retention cohorts
— 05_ab_test_analysis.sql	# Recommendation experiment
— notebooks/	

```
| |— analysis.ipynb          # Main analysis notebook
| |— statistical_tests.ipynb # A/B test statistics
|— visualizations/
| |— dashboard.pbix / .twb  # Interactive dashboard (or)
| |— plots/                # Static PNG exports if no BI tool
|— executive_summary.pdf    # 2-page business report
|— data/                   # Cleaned datasets (optional)
  |— customer_master.csv
```

README Must Include:

- Setup instructions (DB setup, package requirements)
- How to run the analysis end-to-end
- Key findings (3-5 bullet points)
- Recommendations (prioritized list)
- Assumptions made
- Time spent on each section

Bonus Challenges

1. Predictive Churn Model

- Build a logistic regression or simple tree model to predict 90-day churn
- Feature importance analysis
- Identify top 1000 customers at risk

2. Marketing Channel Attribution

- Multi-touch attribution: customers who appear in multiple channels
- Calculate incrementality of each channel

3. Dynamic Pricing Impact

- Analyze how `unit_price` variance affects conversion
- Optimal discount threshold recommendation

4. SQL Optimization Challenge

- Provide EXPLAIN ANALYZE for your slowest query
 - Optimize it to run 3x faster
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